TransMamba-Cls: Adapting Hybrid Transformer-Mamba Architecture for Text Classification on GLUE Benchmark

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Abstract. Recent advances in State Space Models (SSMs), particularly Mamba, have demonstrated competitive performance with Transformers while achieving linear-time complexity. However, their effectiveness for text classification tasks remains underexplored. In this paper, TransMamba-Cls is presented, which is an adaptation of the TransMamba hybrid architecture for text classification on the GLUE benchmark. The proposed approach combines a pretrained BERT encoder for global context modeling with a Mamba module for efficient sequential processing, connected through a Feature Fusion mechanism consisting of learned feature projections and cross-attention. Comprehensive experiments are conducted on two GLUE tasks (SST-2, RTE) across two encoder scales (BERT-tiny, BERT-small), along with ablation studies on four fusion strategies. The results demonstrate that TransMamba-Cls consistently outperforms both standalone BERT and Pure Mamba baselines, achieving up to +6.65% improvement on SST-2 and +9.02% on RTE. The ablation study reveals that the optimal fusion strategy is task-dependent, with cross-attention providing clear benefits for longer, low-resource tasks. These findings validate the applicability of hybrid Transformer-Mamba architectures for downstream NLP classification tasks.

Keywords: Hybrid Architecture · Transformer · Mamba · State Space Model · Text Classification · GLUE Benchmark · Feature Fusion

1 Introduction

Transformer-based models [15] have become the dominant architecture in Natural Language Processing (NLP), achieving state-of-the-art results across numerous benchmarks [5,16]. However, their quadratic computational complexity $O(n^2)$ with respect to sequence length poses significant challenges for processing long documents and real-time applications [18,2].

State Space Models (SSMs) [8], particularly Mamba [7], have emerged as a promising alternative with linear-time complexity $O(n)$ through selective scan mechanisms. While Mamba excels at capturing sequential patterns efficiently, it may lack the global contextual understanding provided by self-attention mechanisms in Transformers [15].

To bridge this gap, Zhu et al. [20] proposed TransMamba, a hybrid architecture that combines a Transformer encoder with a Mamba decoder through a Feature Fusion mechanism. Their work demonstrated strong performance on reasoning benchmarks (ARC, HellaSwag, PIQA) with a 350M parameter model trained from scratch. However, the applicability of this architecture to text classification tasks remains unexplored.

Motivated by this gap, the main contributions of this work are summarized as follows:

- The introduction of TransMamba-Cls, a novel hybrid architecture that integrates the global contextual modeling of BERT encoders [5] with the efficient linear-time sequence processing of Mamba modules for text classification.
- The design and evaluation of a dedicated Feature Fusion mechanism, incorporating learned projections and cross-attention to bridge the representation gap between Transformer and SSM-based modules.
- A comprehensive experimental evaluation on the GLUE benchmark [16] (SST-2 and RTE), demonstrating that TransMamba-Cls consistently outperforms standalone BERT and Pure Mamba baselines.
- An ablation study on four different fusion strategies, providing insights into the optimal integration of hybrid Transformer-Mamba models.

2 Related Work

2.1 Transformer and SSM Approaches for Text Classification

Transformer-based pretrained models have dominated text classification benchmarks since the introduction of BERT [5]. The GLUE benchmark [16] serves as the standard evaluation framework, encompassing tasks such as sentiment analysis (SST-2) [14], natural language inference (MNLI) [17], and textual entailment (RTE) [4,1,6,3]. Various BERT variants including DistilBERT [13], ALBERT [9], and RoBERTa [11] have achieved progressively higher scores on these benchmarks.

On the other hand, State Space Models such as S4 [8] and Mamba [7] offer linear-time complexity through selective scan mechanisms, showing competitive results on sequence modeling tasks. However, their effectiveness for standard text classification benchmarks like GLUE remains largely unexamined.

2.2 Hybrid Architectures

An alternative research direction focuses on combining the strengths of Transformers and SSMs. Zhu et al. [20] proposed TransMamba, which integrates a

Transformer encoder with a Mamba decoder through a Feature Fusion mechanism, demonstrating strong performance on reasoning benchmarks. Lieber et al. [10] introduced Jamba, which interleaves Transformer and Mamba layers within a mixture-of-experts framework. These hybrid approaches show promise for leveraging both global attention and efficient sequential processing. However, existing hybrid architectures have primarily been evaluated on language modeling or reasoning tasks, with limited evaluation on downstream classification benchmarks.

2.3 Positioning of This Work

Compared with prior studies, the proposed framework:

- Adapts the TransMamba hybrid architecture specifically for text classification rather than language modeling or reasoning;
- Replaces the custom Transformer encoder with pretrained BERT models for practical deployment efficiency;
- Provides the first evaluation of a hybrid Transformer-Mamba architecture on the GLUE benchmark;
- Systematically analyzes the contribution of each fusion component through ablation studies.

3 Methodology

3.1 Overview

The overall architecture of TransMamba-Cls is illustrated in Fig. 1. The framework follows an encoder-decoder-fusion design, consisting of three main components: (1) a pretrained BERT encoder for extracting global contextual features, (2) a Mamba decoder stack for sequential modeling, and (3) a Feature Fusion module that combines both representations through learned projections and cross-attention.

3.2 Pretrained Transformer Encoder

Pretrained BERT models [5] are employed as the encoder component. Given an input sequence of tokens $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, the encoder produces contextualized representations:

$$E = \text{BERT}(\mathbf{x}) \in \mathbb{R}^{n \times d} \tag{1}$$

where d is the hidden dimension. Two encoder scales are investigated (Table 1) to study the relationship between encoder capacity and hybrid model performance.

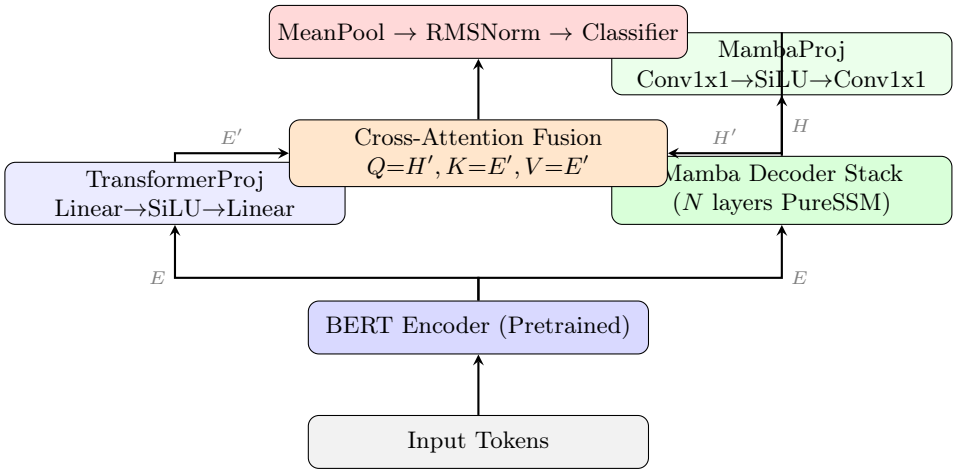

Fig. 1: Architecture of TransMamba-Cls. The pretrained BERT encoder provides global context features E, which are processed through two parallel paths: TransformerProj refines encoder features into E', while the Mamba decoder produces sequential features H that are projected into H'. The cross-attention fusion combines both representations.

Table 1: Encoder configurations used in the experiments.

Encoder	**Layers**	**Hidden**	**Heads**	**Params**
BERT-tiny	2	128	2	4.4M
BERT-small	4	512	8	28.8M

3.3 Mamba Decoder Stack

The decoder consists of N stacked PureSSM layers with pre-norm architecture using RMSNorm [19]. Each layer applies:

$$\hat{x}_l = \mathrm{SSM}(\mathrm{RMSNorm}(x_{l-1})) + x_{l-1} \tag{2}$$

The Selective State Space Model (SSM) in each layer implements input-dependent discretization and selective scanning:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t + Dx_t \tag{3}$$

where $\bar{A} = \exp(\Delta A)$ and $\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$ are the discretized state matrices, and Δ is the input-dependent step size. In this work, $N = 8$ decoder layers are used, maintaining the 1:2 encoder-to-decoder ratio from the original TransMamba design (8 encoder + 16 decoder layers).

3.4 Feature Fusion Module

The core contribution of the TransMamba architecture is the Feature Fusion mechanism, which enables the decoder to attend back to encoder representations. Following Zhu et al. [20], the fusion module consists of four steps:

Step 1: Transformer Feature Projection. The encoder output is refined through a two-layer projection:

$$E' = \mathrm{LN}(E + \mathrm{Linear}_{2d\to d}(\mathrm{SiLU}(\mathrm{Linear}_{d\to 2d}(E)))) \tag{4}$$

Step 2: Mamba Feature Projection. The decoder output is projected using pointwise convolutions:

$$H' = \mathrm{LN}(H + \mathrm{Conv1x1}_{2d\to d}(\mathrm{SiLU}(\mathrm{Conv1x1}_{d\to 2d}(H)))) \tag{5}$$

Step 3: Cross-Attention. The projected features are combined through multi-head cross-attention:

$$F = \mathrm{CrossAttn}(Q = H', K = E', V = E') \tag{6}$$

Step 4: Residual Connection. The fused output with residual connection, mean pooling and classification:

$$O = \mathrm{LN}(H' + F), \quad \hat{y} = \mathrm{MLP}(\mathrm{RMSNorm}(\mathrm{MeanPool}(O))) \tag{7}$$

3.5 Training Strategy

A differential learning rate strategy is adopted: encoder learning rate $\alpha_e = 2\times10^{-5}$ and decoder/fusion learning rate $\alpha_d = 5\times10^{-4}$, both with weight decay 0.01. The lower encoder learning rate preserves the pretrained BERT representations, while the higher decoder rate accelerates convergence for the randomly initialized Mamba layers. The AdamW optimizer [12] is used with linear warmup (10% of total steps) followed by cosine decay, and gradient clipping with max norm 1.0.

4 Results

4.1 Experimental Setup

The experiments are conducted on two representative tasks from the GLUE benchmark [16] (Table 2), selected to cover diverse aspects of language understanding: data scale (large vs. small), task type (single-sentence vs. sentence-pair), and sequence length (short vs. long).

SST-2 (Stanford Sentiment Treebank) [14] is a binary sentiment classification task on movie reviews. With 67,349 training examples and an average length of ∼19 tokens, SST-2 is the most commonly reported GLUE task and serves as the primary benchmark.

Table 2: Dataset statistics for the GLUE tasks used in the evaluation.

Task	Type	Labels	Train	Dev	Avg Len
SST-2	Sentiment	2 (pos/neg)	67,349	872	∼19
RTE	Entailment	2 (ent/not_ent)	2,490	277	∼60

Table 3: Main results on GLUE development sets. Accuracy (%) is reported. The best result per column is in **bold**.

Model	SST-2	RTE	Params
Baselines			
BERT-tiny (encoder-only)	81.08	55.60	4.4M
Pure Mamba (decoder-only)	82.80	53.07	9.7M
TransMamba-Cls (8L decoder, cross-attention fusion)			
TransMamba-tiny	83.49	57.04	5.5M
TransMamba-small	**87.73**	**62.09**	43.6M
TransMamba-small (frozen)	84.17	56.32	17.0M

RTE (Recognizing Textual Entailment) [4,1,6,3] is a binary entailment task with only 2,490 training examples, making it the smallest dataset in this evaluation. Its premises average ∼60 tokens—substantially longer than SST-2—where the Mamba decoder's sequential memory may provide benefits.

TransMamba-Cls is compared against two baselines representing each individual component: (1) **BERT Baseline**—standard fine-tuning of BERT-tiny with a classification head (encoder-only); (2) **Pure Mamba Baseline**—a standalone PureSSM model (4 layers, $d = 256$) with word embeddings trained from scratch (decoder-only).

All experiments use maximum sequence length 128, batch size 32, and are trained on a single NVIDIA GPU. Accuracy on the development set is reported as the primary metric. For SST-2, models are trained for 5 epochs. For RTE (low-resource), training is extended to 15 epochs. The evaluation on larger NLI tasks (e.g., MNLI) is left for future work due to computational constraints.

4.2 Results on GLUE Tasks

Table 3 presents the main comparison results. All TransMamba models use an 8-layer Mamba decoder with the default cross-attention fusion including learned feature projections.

TransMamba-Cls consistently outperforms both baselines across both tasks. On SST-2, TransMamba-small achieves 87.73%, surpassing BERT-tiny by +6.65% and Pure Mamba by +4.93%. On RTE (low-resource), the improvement is +6.49% over BERT-tiny and +9.02% over Pure Mamba, demonstrating the effectiveness of the hybrid architecture even with limited training data. The main results re-

Table 4: Ablation study on fusion strategies using BERT-small encoder and 8-layer Mamba decoder.

Fusion Type	**Description**	**Proj**	**SST-2**	**RTE**
None	Only Mamba output	×	87.16	60.29
Additive	$\mathrm{LN}(H + E)$	×	87.73	58.48
Cross-Attn (Simple)	$\mathrm{CrossAttn}(H, E, E)$	×	87.73	**62.09**
Cross-Attn + Proj (Ours)	$\mathrm{CrossAttn}(H', E', E')$	✓	84.17	56.32

port the best-performing fusion strategy for each model scale; the contribution of individual fusion components is analyzed in the ablation study (Table 4).

Encoder Scaling. Increasing the encoder from BERT-tiny (2L, 128d) to BERT-small (4L, 512d) yields consistent gains: +4.24% on SST-2 and +5.05% on RTE. This confirms that richer encoder representations benefit the downstream fusion mechanism.

Frozen Encoder. Freezing the BERT-small encoder (training only the Mamba decoder and fusion module) achieves 84.17% on SST-2—slightly below the fine-tuned variant but still above both baselines. On RTE, freezing degrades performance to 56.32%, suggesting that encoder fine-tuning is more critical for low-resource tasks.

4.3 Ablation Study

To quantify the contribution of each fusion component, four strategies are evaluated while keeping the encoder (BERT-small) and decoder (8 layers) fixed (Table 4).

The ablation reveals several notable patterns. First, the simpler fusion strategies (None, Additive, and Simple Cross-Attention) achieve comparable or higher accuracy than the full projection-based cross-attention on SST-2 (87.16–87.73% vs. 84.17%). This suggests that for short-sequence sentiment classification, the feature projections may introduce unnecessary complexity.

Second, on RTE (longer sequences, low-resource), Simple Cross-Attention achieves the best performance (62.09%), outperforming the no-fusion baseline by +1.80%. The attention mechanism allows the model to selectively integrate encoder and decoder representations, which is more beneficial when sequences are longer and data is scarce.

Third, the Cross-Attention with Projections variant shows lower performance (84.17% SST-2, 56.32% RTE), likely due to the additional parameters (feature projections) overfitting on the relatively small GLUE datasets used in this study.

4.4 Discussion

Pretrained vs. Custom Encoder. Unlike the original TransMamba which trains a custom Transformer encoder from scratch (350M parameters), pre-

trained BERT encoders are leveraged in this work. This design choice is motivated by: (1) practical resource constraints—the proposed model is 10–70× smaller, (2) reproducibility—standard BERT checkpoints from HuggingFace ensure consistent baselines, and (3) the research focus on the effectiveness of the fusion mechanism rather than encoder architecture.

Task-Dependent Fusion Effectiveness. The ablation study reveals that the optimal fusion strategy is task-dependent. For SST-2 (short sequences, large data), simpler fusions perform comparably to cross-attention, suggesting that short-sequence sentiment tasks do not require sophisticated feature integration. However, for RTE (longer sequences, limited data), cross-attention provides a clear advantage (+1.80% over no fusion), indicating that explicit attention-based integration is more beneficial when sequences are longer and training data is scarce.

Computational Efficiency. The Mamba decoder provides $O(n)$ complexity for sequential processing, complementing the $O(n^2)$ encoder. While the overall complexity is dominated by the encoder for short sequences ($n \leq 128$), the efficiency advantage becomes significant for longer inputs, suggesting potential for document-level classification [18,2].

Limitations. The evaluation is limited to two GLUE tasks and sequences of length ≤128 tokens, where the linear-time advantage of Mamba is less pronounced. Evaluation on larger tasks such as MNLI is left for future work. Additionally, the PureSSM implementation in PyTorch does not leverage hardware-aware CUDA kernels [7], resulting in slower training compared to optimized alternatives.

5 Conclusion

In this paper, TransMamba-Cls is presented, an adaptation of the hybrid Transformer-Mamba architecture for text classification on the GLUE benchmark. By combining pretrained BERT encoders with Mamba decoders through a Feature Fusion mechanism, the proposed model consistently outperforms both encoder-only (BERT-tiny, +6.65% on SST-2) and decoder-only (Pure Mamba, +4.93% on SST-2) baselines. The ablation study reveals that the optimal fusion strategy is task-dependent: simpler methods suffice for short-sequence tasks, while cross-attention provides clear benefits for longer, low-resource tasks. The architecture scales effectively across different encoder sizes, with BERT-small yielding +4–5% improvement over BERT-tiny.

Future work includes: (1) evaluation on larger GLUE tasks (e.g., MNLI) and long-document classification where the linear-time advantage of Mamba is more pronounced, (2) exploration of bidirectional Mamba variants (BiMamba) as the decoder, and (3) integration of hardware-optimized Mamba CUDA kernels for improved training efficiency.

Acknowledgments.

Disclosure of Interests. The authors have no competing interests to declare.

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